A Four-Dimensional $SU(2)$ Quotient–Fiber Block Theorem

From Gauge-Zero-Mode Removal to Local Heat-Kernel Transport

Research dossier—finite theorem, nonlinear local stability, and frontier audit

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Abstract

We construct an explicit four-dimensional 2^4 cubical block for the tangent theory of $SU(2)$ lattice gauge fields and separate infinitesimal gauge modes, retained coarse curvature data, and eliminated conditional-fiber variables. The scalar block has 81 vertices, 216 links, and 216 plaquettes. After removing gauge directions its physical tangent space has dimension 136; the 24 displayed coarse boundary-face fluxes have exact rank 17, leaving a 119-dimensional scalar fiber. Tensoring with $\mathfrak{su}(2)$ gives dimensions 408, 51, and 357. A product-Hodge calculation proves $\|d_1 A\|^2 \geq 2\|A\|^2$ on the gauge-orthogonal quotient and hence on the fiber. Gaussian conditioning changes only the transported mean; the centered fiber covariance is exactly independent of retained data. We then prove a finite-dimensional nonlinear stability theorem: explicit Hessian and centered cross-gradient bounds for the local $SU(2)$ heat-kernel action imply strong convexity and a Wasserstein fiber coefficient below one. Heat-kernel asymptotics and BCH analyticity ensure such bounds on a sufficiently small local tube, but no explicit cutoff-uniform tube, global orbit atlas, defect polymer estimate, multiscale iteration, or infrared crossover is proved.

Claim level. The cubical, Hodge, rank, and Gaussian results are PROVED restricted finite-block theorems. The nonlinear transport implication is PROVED from explicit derivative bounds; its local heat-kernel realization has existential finite-block constants. Full-measure RG iteration, crossover, continuum existence, and the Clay mass gap remain OPEN.

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1 Why this is the next target

The exact Clay problem asks for a nontrivial axiomatic pure Yang–Mills theory on $\mathbb{R}^4$ with a positive Hamiltonian gap for every compact simple gauge group [1]. Earlier stages of the present programme established two diagnostic facts. First, translated concentrated heat kernels have total-variation distance tending to one, so a worst-case microscopic TV basin cannot be cutoff uniform. Second, a block variable can respond to boundary data because it transports a retained coarse mode even when its centered fluctuation is independent of that data. The local object to control is thus the conditional fiber after gauge and retained modes have been removed.

This paper moves that idea from a two-variable abelian calibration to an explicit four-dimensional cubical gauge complex with $\mathfrak{su}(2)$ color. It does not attempt the ultraviolet-to-infrared crossover. The target is the finite local theorem that must precede such an iteration.

We use the heat-kernel plaquette action because it is an exact class-function regulator with transparent weak-field asymptotics and a long-standing role as an alternative lattice action [2]. Background minimizers and gauge-covariant expansions are central in Balaban's RG analysis [3, 4]; our local centering is designed to be compatible with that architecture.

2 The frozen cubical block

Let I_2 be the cellular interval with vertices $\{0, 1, 2\}$ and two positively oriented edges. The block is

$$B = I_2^4.$$

Let $C^q(B;\mathbb{R})$ be the real cellular q-cochains with the standard Euclidean inner product. The first two coboundaries are

$$d_0 : C^0(B;\mathbb{R}) \longrightarrow C^1(B;\mathbb{R}), \qquad d_1 : C^1(B;\mathbb{R}) \longrightarrow C^2(B;\mathbb{R}),$$

with $d_1 d_0 = 0$.

Proposition 2.1 (Cell counts and exactness, PROVED)**.** *The block has*

$$|B_0| = 81, \qquad |B_1| = 216, \qquad |B_2| = 216.$$

Moreover,

$$\operatorname{rank} d_0 = 80, \qquad \operatorname{rank} d_1 = 136, \qquad \ker d_1 = \operatorname{im} d_0.$$

Proof. Product counting gives

$$|B_0| = 3^4, \quad |B_1| = 4 \cdot 2 \cdot 3^3, \quad |B_2| = \binom{4}{2} 2^2 3^2.$$

The one-skeleton is connected, so $\ker d_0$ consists of constants and $\operatorname{rank} d_0 = 81 - 1 = 80$. The cubical block is contractible, hence $H^1(B;\mathbb{R}) = 0$ and $\ker d_1 = \operatorname{im} d_0$. Rank–nullity then gives $\operatorname{rank} d_1 = 216 - 80 = 136$. □

The infinitesimal link field for $SU(2)$ is

$$\mathcal{A} = C^1(B;\mathbb{R}) \otimes \mathfrak{su}(2).$$

The infinitesimal gauge directions are $\operatorname{im} d_0 \otimes \mathfrak{su}(2)$. We represent the quotient by the orthogonal Coulomb slice

$$\mathcal{H} = \ker d_0^* \otimes \mathfrak{su}(2). \tag{1}$$

This is a local tangent representation, not a global gauge fixing.

Corollary 2.2 (Physical tangent dimension, PROVED)**.** *The scalar quotient* $\ker d_0^*$ *has dimension* 136*. The* $\mathfrak{su}(2)$*-valued physical tangent space* $\mathcal{H}$ *has dimension* 408*, while the gauge subspace has dimension* 240*.*

3 Retained coarse curvature and the conditional fiber

For every coordinate pair $\mu < \nu$ and every choice of the remaining two coordinates in $\{0, 2\}$, sum $d_1 A$ over the four fine plaquettes in that 2×2 boundary face. This defines

$$C : C^1(B;\mathbb{R}) \longrightarrow \mathbb{R}^{24}.$$

It is invariant under linearized gauge transformations because $C d_0 = 0$.

Theorem 3.1 (Exact retained rank, PROVED)**.** *The* 24 *displayed coarse face fluxes have rank* 17*. Their restriction to* $\ker d_0^*$ *also has rank* 17*.*

Proof. Let $P : C^1(I_2^4) \to C^1(I_1^4)$ sum each pair of fine boundary links into a coarse link. Discrete Stokes gives $C = d_{1,\mathrm{coarse}} P$. The map P is onto. The single coarse hypercube has 16 vertices and 32 links and is contractible, so

$$\operatorname{rank} d_{1,\mathrm{coarse}} = 32 - (16 - 1) = 17.$$

Therefore $\operatorname{rank} C = 17$. Since C vanishes on $\operatorname{im} d_0$ and $C^1 = \operatorname{im} d_0 \oplus \ker d_0^*$, its restriction to the orthogonal quotient has the same rank. The accompanying exact rational elimination independently certifies the rank. □

Choose an orthonormal identification of the image of $C|_{\ker d_0^*}$ with $\mathbb{R}^{17}$ and denote the resulting independent retained map by

$$\overline{C} : \ker d_0^* \to \mathbb{R}^{17}.$$

Define the scalar fiber and the $\mathfrak{su}(2)$ fiber by

$$F = \ker d_0^* \cap \ker C, \qquad \mathcal{F} = F \otimes \mathfrak{su}(2). \tag{2}$$

Corollary 3.2 (Base and fiber dimensions, PROVED)**.** *The scalar base and fiber dimensions are* 17 *and* 119*. Their* $\mathfrak{su}(2)$ *counterparts have dimensions*

$$\dim \mathcal{Y} = 51, \qquad \dim \mathcal{F} = 357.$$

The seven dependencies among the 24 displayed fluxes are structural coarse Bianchi/boundary relations. Treating all 24 as independent would make the conditional problem ill-typed.

4 Certified quotient-fiber coercivity

The tangent Yang–Mills quadratic form is

$$Q_t(A) = \frac{1}{2t}\|d_1 A\|^2, \qquad A \in \ker d_0^*.$$

Theorem 4.1 (Product-Hodge coercivity, PROVED)**.** *For the block I_2^4,*

$$\|d_1 A\|^2 \geq 2\|A\|^2 \qquad (A \in \ker d_0^*). \tag{3}$$

Consequently the same inequality holds for $A \in F$ and componentwise on $\mathcal{F}$.

Proof. On the interval I_2, the nonzero spectra of the vertex and edge Laplacians are

$$\operatorname{spec}(d_0^* d_0) = \{0, 1, 3\}, \qquad \operatorname{spec}(d_0 d_0^*) = \{1, 3\}.$$

The Hodge Laplacian on product one-cochains is the tensor sum of one edge Laplacian and three vertex Laplacians. Its eigenvalue-one sector uses the edge eigenvalue 1 and three constant vertex modes. These modes are exact: they are gradients of the corresponding vertex eigenfunctions. Hence they are orthogonal to $\ker d_0^*$. The next possible tensor-sum eigenvalue is $1 + 1 = 2$. On $\ker d_0^*$,

$$\Delta_1 = d_0 d_0^* + d_1^* d_1 = d_1^* d_1,$$

which proves (3). Restriction to F can only increase the Rayleigh minimum, and tensoring with $\mathfrak{su}(2)$ repeats each scalar eigenvalue three times. □

The executable spectral calculation gives

$$\lambda_{\min}(d_1^* d_1|_{\ker d_0^*}) = 2, \qquad \lambda_{\min}(d_1^* d_1|_F) \approx 3.438447187.$$

The theorem uses the analytic lower bound 2; the sharper number is a floating-point calibration.

5 Exact Gaussian conditional-fiber theorem

Let $L = d_1^* d_1|_{\ker d_0^*}$ and choose independent base coordinates so that $B : \ker d_0^* \to \mathbb{R}^{17}$ has full row rank. Consider the centered Gaussian tangent measure with precision L/t, conditioned on $BA = y$.

Theorem 5.1 (Transported mean and invariant centered fiber, PROVED)**.** *Let*

$$G = BL^{-1}B^*, \qquad M = L^{-1}B^*G^{-1}.$$

Then

$$\mathbb{E}[A \mid BA = y] = My, \tag{4}$$

$$\operatorname{Cov}(A \mid BA = y) = t\big[L^{-1} - L^{-1}B^*G^{-1}BL^{-1}\big]. \tag{5}$$

In particular, the centered conditional variable $Z = A - My$ has a law independent of y and is supported on F. Its centered base-response coefficient is exactly zero.

Proof. The matrix G is positive definite because B has full row rank and $L > 0$. The standard constrained-Gaussian completion of the square gives (4)–(5). Neither the covariance nor its support depends on y. Subtracting My therefore produces the same fiber law for every retained value. □

The computed Euclidean operator norm of M is approximately 0.393174006. This number depends on the normalization of base coordinates and is not a physical invariant. The exact zero after centering is the invariant content.

6 Local nonlinear $SU(2)$ heat-kernel stability

Let p_t be the $SU(2)$ heat kernel and let $U_e = \exp A_e$ in a sufficiently small logarithmic chart. The local block density, including the logarithmic Haar Jacobian $J(A)$, has action

$$S_t(A) = -\sum_{p \subset B} \log p_t(U_p(A)) - \log J(A). \tag{6}$$

On a local Coulomb slice, parallel transport all coarse boundary-face holonomies to a fixed root and take logarithms. At the trivial connection the differential of this displayed retained map is $C \otimes I_{\mathfrak{su}(2)}$ and has rank 51. Choose 51 fixed independent linear combinations whose differential is surjective. The submersion theorem then gives local coordinates

$$A = \Phi(y, z), qquady \in \mathbb{R}^{51}, \quad z \in \mathbb{R}^{357},$$

in which y is retained and z parameterizes the conditional fiber.

Lemma 6.1 (Finite-block heat-kernel expansion, PROVED (local))**.** *There exist $t_0, \rho_0, C_B, c_B > 0$ such that, for $0 < t \le t_0$ and $\|\Phi(y,z)\|_\infty \le \rho \le \rho_0$,*

$$\left\| t\nabla_z^2 S_t(\Phi(y,z)) - L_F \otimes I_{\mathfrak{su}(2)} \right\| \le C_B\big(\rho + t + e^{-c_B/t}\big), \tag{7}$$

after a fixed bounded normalization of the local coordinates. Here $L_F = d_1^* d_1|_F$ at the origin.

Proof. On a compact logarithmic ball strictly inside the cut locus, the heat-kernel parametrix and all of its required derivatives have the form

$$-\log p_t(\exp X) = \frac{\|X\|^2}{2t} + a_0(X) + O(e^{-c/t}),$$

after reducing c if derivatives contribute polynomial powers of t^{-1}. The functions a_0 and the logarithmic Haar Jacobian have bounded derivatives. The plaquette BCH expansion is

$$\log U_p(A) = d_1 A|_p + O(\|A\|^2),$$

with bounded derivatives on the fixed finite block. Differentiating twice shows that the difference from $t^{-1} d_1^* d_1$ is $O(\rho/t) + O(1) + O(e^{-c/t}/t)$. Multiplication by t yields (7). The local coordinate map and its inverse have bounded derivatives after shrinking ρ_0. □

Theorem 6.2 (Nonlinear fiber coercivity, PROVED)**.** *Set*

$$\eta = C_B\big(\rho + t + e^{-c_B/t}\big).$$

Whenever $\eta < 2$, the local nonlinear fiber potential is strongly convex with

$$\nabla_z^2 S_t(\Phi(y,z)) \ge \frac{2-\eta}{t} I. \tag{8}$$

It consequently has a unique conditional minimizer in every convex local fiber domain on which the bound holds.

Proof. Theorem 4.1 gives $L_F \otimes I_{\mathfrak{su}(2)} \ge 2I$. Apply the operator-norm perturbation bound (7). Strong convexity gives uniqueness. □

This is a genuine local nonlinear statement, but the constants are existential finite-block constants. A multiscale theorem requires explicit values in an effective-action norm.

7 Centered transport from derivative certificates

Let $z_t^\star(y)$ be the conditional minimizer and recenter the common convex fiber chart $D \subset \mathbb{R}^{357}$ by $w = z - z_t^\star(y)$. Write the centered potential as $V_y(w)$ and the normalized measure as

$$\widetilde{\nu}_y(dw) = Z_y^{-1} e^{-V_y(w)} \mathbf{1}_D(w)\, dw.$$

Lemma 7.1 (Strong-convexity transport, PROVED)**.** *Suppose $\nabla^2 V_y \geq \alpha I$ on the same convex domain D for all y, and*

$$\sup_{w \in D} \|\nabla V_y(w) - \nabla V_{y'}(w)\| \leq L\, d_Y(y, y').$$

Then

$$W_2(\widetilde{\nu}_y, \widetilde{\nu}_{y'}) \leq \frac{L}{\alpha} d_Y(y, y'). \tag{9}$$

Proof. Strong convexity on a convex domain gives the logarithmic Sobolev inequality with constant $1/\alpha$. Applied to the density ratio, it bounds relative entropy by $1/(2\alpha)$ times relative Fisher information. The gradient hypothesis bounds that Fisher information by $L^2 d_Y(y, y')^2$. The logarithmic Sobolev inequality implies the quadratic transport inequality [5], yielding (9). □

For the quadratic tangent action, centering makes $L = 0$.

Lemma 7.2 (Centered cross-gradient cancellation, PROVED (local))**.** *The local coordinates may be chosen so that their tangent splitting at the origin is the Gaussian conditional-mean direction plus the fiber $\mathcal{F}$. There then exist $C'_B, t_1, \rho_1 > 0$ such that, after centering at the nonlinear conditional minimizer and restricting to a common convex fiber ball,*

$$t \sup_{w \in D} \|\nabla V_y(w) - \nabla V_{y'}(w)\| \leq \chi\, d_Y(y, y'), \qquad \chi \leq C'_B \rho. \tag{10}$$

for $0 < t \leq t_1$ and a sufficiently small tube of radius $\rho \leq \rho_1$.

Proof. Choose the horizontal derivative of Φ at the origin to be the L-orthogonal conditional-mean map M from Theorem 5.1; choose its vertical derivative to identify $\mathbb{R}^{357}$ with $\mathcal{F}$. The mixed base–fiber Hessian of the leading quadratic action then vanishes. Centering at the nonlinear minimizer also makes the fiber gradient vanish at $w = 0$. After subtracting the quadratic heat-kernel term, the map

$$(t, y, w) \longmapsto t \nabla_w V_y(w)$$

has uniformly bounded derivatives on a smaller local tube by the same parametrix and BCH argument as Lemma 6.1; exponentially small image terms are absorbed by decreasing t_1. Its base derivative at the origin is zero. The mean-value theorem therefore bounds its base derivative by $C'_B \rho$, proving (10). □

The factor ρ records that the leading Gaussian cross-response has been removed; what remains begins with nonlinear commutator and coordinate terms.

Corollary 7.3 (Local centered-fiber contraction, PROVED (local))**.** *If* (7) *and* (10) *hold with*

$$K_{\text{fiber}} := \frac{\chi}{2 - \eta} < 1,$$

then

$$W_2(\widetilde{\nu}_y, \widetilde{\nu}_{y'}) \leq K_{\text{fiber}} d_Y(y, y').$$

For the fixed block, sufficiently small ρ and t make this inequality true with existential constants.

Proof. Use $\alpha = (2-\eta)/t$ from Theorem 6.2 and $L = \chi/t$ from Lemma 7.2 in Lemma 7.1. Since $\eta \to 0$ and $\chi \to 0$ as $(\rho, t) \to (0,0)$ inside the local tube, shrink the tube until $\chi/(2-\eta) < 1$. □

The common-domain restriction is deliberate. The probability that leaves this good tube belongs to the defect channel; it cannot be discarded.

8 Executable calibration and hostile tests

The accompanying implementation constructs d_0, d_1, C, uses exact rational elimination for the retained rank, and computes orthogonal quotient and fiber bases numerically. Its output is:

Quantity	Value
scalar gauge rank	80
scalar physical dimension	136
displayed / independent coarse fluxes	24/17
scalar fiber dimension	119
$\mathfrak{su}(2)$ physical / base / fiber dimensions	408/51/357
physical eigenvalue range	$[2, 12]$
fiber eigenvalue range	$[3.438447187, 12]$
minimum nonzero coarse singular value	2.828427125
full conditional-mean response norm	0.393174006
centered Gaussian response	0

Eleven automated tests include four hostile checks:

1. without quotienting, the quadratic Hessian has exactly 80 scalar gauge zero modes;
2. the 24 displayed retained fluxes are not falsely treated as independent;
3. invalid geometries and base dimensions fail closed;
4. a nonlinear remainder that exhausts the certified margin is rejected.

The numerical eigenvalue and response norm calibrate the theorem; only the analytic lower bound 2, exact ranks, and Gaussian identities enter proved claims.

9 RG certificate and large-field ledger

A complete block morphism would transport

$$(\mu_k, S_k, \mathcal{O}_k, \ell_k) \longmapsto (\mu_{k+1}, S_{k+1}, \mathcal{O}_{k+1}, b\ell_k)$$

together with the retained map, conditional fiber, local interaction norm, running coupling, reconstruction error, and positivity certificate. The present package supplies only the finite local tangent and good-tube entries.

ID	Required result	Status
F1	Replace the local Coulomb/constant-rank chart by an exact gauge-covariant nonlinear disintegration, with orbit-atlas overlaps and stabilizers.	OPEN
F2	Derive explicit C_B, C'_B, ρ_0, t_0 and prove stability for the full generated effective-action class under one RG step.	OPEN
F3	Bound the complement of the good tube by a dependent polymer norm, including cut-locus configurations, reducible orbits, center flux, global holonomy, and topological sectors.	OPEN
F4	Iterate through $k(a) \sim \log_b(\ell_*/a) \to \infty$ ultraviolet steps with summable errors and controlled observable support.	OPEN
F5	Prove that the retained base crosses into a massive basin at a finite physical scale.	OPEN
F6	Construct nontrivial regulator-independent continuum Schwinger functions, verify the OS hypotheses and observable density, and transfer decay to the Hamiltonian spectrum.	OPEN

Reflection positivity need not be manifest in every intermediate coordinate system if the RG integration is exact and the original regulated observables are transported without error; otherwise it requires a quantitative recovery certificate. No such multiscale certificate is supplied here.

10 What has and has not advanced

The result establishes that the retained/fiber distinction survives the first genuinely four-dimensional nonabelian-compatible test. Gauge directions are removed exactly at tangent level, retained curvature data have the correct rank, the fiber has a certified coercivity margin, and nonlinear heat-kernel geometry preserves that margin locally. The centered transport coefficient is therefore a mathematically viable local target rather than a purely ontological proposal.

The result does not generate infrared mass. Fiber contraction controls the degrees of freedom discarded by blocking; the retained base can still carry massless long-wavelength modes. The central Yang–Mills problem is now located more sharply in F4–F5: control a diverging RG trajectory and prove a nonperturbative crossover of the base into a massive regime. The local theorem is an input to that task, not its conclusion.

11 Immediate next theorem

The next bounded theorem should attach the certificate to a Balaban-type effective-action norm. For one exact RG step, prove

$$t_k \|\nabla_F^2 (S_k - Q_k)\| \le \eta_k < 2, \qquad t_k \|\nabla_F V_{k,y} - \nabla_F V_{k,y'}\| \le \chi_k d_Y(y, y'),$$

with

$$\frac{\chi_k}{2 - \eta_k} < 1,$$

and a gauge-invariant defect activity satisfying a declared polymer norm. Every constant must be expressed in variables propagated by the RG map. A failure of this inequality would identify whether the obstruction lies in the chosen retained data, the orbit chart, or the generated interactions.

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